

How can the novelty p3 ERP be distinguished from possible confound? What are the confound?

Below is a concise but technically solid explanation suitable for ERP/BCI/cognitive-neuroscience work.

Distinguishing the Novelty P3 from Confounds

The **Novelty P3** (also called **P3a**) is an **involuntary fronto-central ERP positivity**, typically peaking around **250–350 ms**, elicited by **unexpected, rare, salient, or novel stimuli** in an oddball sequence. Its defining feature is **automatic orienting** rather than task-relevant evaluation.

To clearly identify a true Novelty P3, you need to distinguish it from several **ERP components** and **task-related confounds** that can mimic or contaminate it.

1. Confounds That Can Mimic a Novelty P3

(A) MMN (Mismatch Negativity)

- **Timing:** 100–200 ms
- **Polarity:** Negative
- **Scalp:** Frontocentral
- **Confound:** A large MMN can create the impression of a later positivity because the baseline shifts.
- **Solution:** Use difference waves, isolate MMN via polarity and time window.

(B) Target-related P3b

- **Timing:** 300–600 ms
- **Scalp:** Parietal maximal
- **Confound:** If novel stimuli accidentally become task-relevant or predictable, they elicit P3b instead of P3a.

- **Solution:** Use **true task irrelevance**, make subjects ignore novel events, **check scalp topography**.
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(C) Stimulus physical differences

Novel stimuli are usually **acoustically or visually different** from standards and deviants.

Confounds include:

- Intensity differences (loudness → larger P3)
- Spectral complexity
- Duration differences
- Sudden onset / sharp transients

Solution: Use physically matched control deviant or equip novelty sounds for loudness, duration, envelope.

(D) Emotional salience

Highly emotional novel sounds (e.g., baby cry, scream) produce a **Late Positive Potential (LPP)** overlapping with the P3a.

This is a **motivational significance** confound.

Solution: Use neutral novel stimuli or measure/control arousal ratings.

(E) Habituation / repetition suppression

If novel stimuli repeat too frequently they are no longer truly novel, and amplitude declines.

This creates a **pseudo-novelty P3** early in each block but not later.

Solution: Use single presentations per novel, or force rarity < 5–10%.

(F) Contingency learning

If a rare event predicts anything (even timing, feedback, or block change), it may elicit P3b rather than P3a.

Solution: Ensure **novel events do not predict reward, timing, or task changes.**

(G) Overlapping motor or oculomotor activity

- Eye blinks/eye movements around 250–350 ms mimic FC positivity.
 - Startle reflex or head movement from unexpected sounds can produce artifacts.
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(H) Post-auricular muscle reflex (PAM)

Auditory stimuli, especially sudden ones, trigger PAM at ~12–18 ms and secondary EMG activity that can contaminate later windows.

2. How to Establish the Novelty P3 Properly

(1) Scalp Topography

- Novelty P3 (P3a): **Fronto-central maximum**
- Target P3b: **Parietal maximum**
- MMN: **Negative peak**, FC distribution

Topography alone is one of the strongest distinguishing features.

(2) Time-window isolation

Typical windows:

- **MMN: 100–200 ms**
- **P3a/Novelty P3: 220–350 ms**
- **P3b: 350–600 ms**

Apply temporal PCA/ICA or cluster-based permutation tests to isolate components.

(3) Orthogonal task design

- Make **novel sounds task-irrelevant**
- Make **targets different** from novel stimuli

- Ensure **novelty does not predict any action**
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(4) Physically matched control deviations

Include **control deviant stimuli** matched in basic acoustics but lacking novelty to check whether P3a reflects true novelty rather than physical salience.

(5) Statistical modeling

Multivariate approaches (e.g., linear mixed models, mass univariate with cluster controls) help test:

- Novel > standard
 - Novel > deviant
 - FC positivity independent of parietal P3b
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(6) Source Localization

Novelty P3 often originates from:

- **Anterior cingulate cortex (ACC)**
- **Inferior frontal gyrus**
- **Insula**

Whereas P3b involves parietal cortex and temporoparietal junction (TPJ).

Source separation helps distinguish the components if discussed in a paper.

3. Summary Table

Component / Confound	Latency	Topography	Source	Confound Mechanism
MMN	100–200 ms	FC	Auditory cortex	Baseline shifts, overlap

Component / Confound	Latency	Topography	Source	Confound Mechanism
P3b	300–600 ms	Parietal	Parietal cortex	Task-relevance accidentally induced
LPP/Emotional	300–800 ms	Centro-parietal	Limbic/parietal	Emotional content of novel sounds
Physical stimulus differences	variable	FC/P	Sensory	Physical salience rather than cognitive novelty
Habituation	trial-dependent	FC	—	Apparent novelty only early in block
Motor/ocular artifacts	0–500 ms	FC	muscles	False positivity
Startle/PAM	early EMG	mastoid/temporal	muscles	Spread into ERP

If you want, I can tailor this to your **EEG design**, provide **confound-free paradigms**, or show **analysis pipelines** for isolating P3a from MMN and P3b.



Can you compare p3a vs lpp for rare emotionally salient sounds?

Below is a clear, research-grade comparison of **P3a vs LPP** specifically for **rare emotionally salient sounds** (e.g., screams, baby cries, **personalized notifications** with emotional value, etc.). This distinction is crucial because **emotional salience and novelty both produce large positivities in overlapping time windows**.

P3a vs LPP for Rare Emotionally Salient Sounds

Feature	P3a (Novelty P3)	LPP (Late Positive Potential)
Core meaning	Automatic orienting to novelty / unexpectedness	Emotional–motivational significance, affective evaluation
Typical latency	220–350 ms post-stimulus	400–800+ ms (sometimes starting ~350 ms)
Scalp distribution	Fronto-central maximal (FCz, Fz most prominent)	Centro-parietal (CPz, Pz, posterior sites)
Triggers	Rare, unexpected, contextually deviant, <i>non-emotional</i> stimuli (scratches, environmental sounds)	High-arousal emotional sounds (screams, cries, threat, reward-related tones)
Task relevance	Not required; P3a is automatic	Stronger when emotional significance is motivationally relevant
Habituation sensitivity	Declines quickly with repetition	Sustained across many repetitions
Dependence on acoustic features	Moderately influenced (intensity, novelty, spectral complexity)	Less driven by physical salience; more by affective meaning
Associated neural generators	ACC, inferior frontal gyrus, anterior insula (orienting network)	Amygdala → parietal/centro-parietal cortices (motivational systems)
Overlaps	250–350 ms positivity	350–800 ms broad positivity
Best dissociation method	Time window + fronto-central topography	Later time window + parietal topography

When Rare Emotional Sounds Produce Both

Rare emotional sounds (e.g., a sudden scream) often elicit **both**:

1. **P3a** (because they are *rare and unexpected*)
2. **LPP** (because they are *emotionally arousing*)

This results in a **biphasic FC→P waveform**:

- Early fronto-central positivity (~250–350 ms) → **P3a**
- Later centro-parietal positivity (~400–800+ ms) → **LPP**

How to Distinguish Them in Your Data

1. Time-window based separation

- P3a: 220–350 ms
- LPP: 350–800 ms (sometimes 400–900)

You can apply temporal PCA/ICA or mass-univariate time-cluster tests to isolate the two windows.

2. Scalp topography

Even when amplitudes are large:

- **P3a: fronto-central**
- **LPP: centro-parietal**

Plotting topomaps for 250–350 ms and 400–800 ms usually reveals the difference clearly.

3. Stimulus design control

If the question is about whether emotion or novelty drives the effect:

Include 3 conditions:

1. Standard
2. Novel neutral
3. Novel emotional

Then:

- **Novel Neutral > Standard** isolates P3a (pure novelty orienting)

- **Novel Emotional > Novel Neutral** isolates emotional LPP

This is the cleanest experimental dissociation.

4. Repetition sensitivity

Emotion-driven LPP is *resistant to habituation*, so if amplitude stays high even after many repetitions, that points to LPP more than P3a.

P3a typically decays rapidly when the same event type is repeated.

5. Source localization

If your pipeline allows sLORETA or minimum-norm:

- P3a: anterior cingulate + frontal salience network
- LPP: parietal areas, sometimes insula–amygdala interactions

Topographic PCA commonly reveals a two-component solution.

Interpretational Framework

If the effect is early, FC-max, and novelty-driven → P3a.

This reflects **automatic orienting to unexpected events**, independent of emotional meaning.

If the effect is late, CP/P-max, and emotion-driven → LPP.

This reflects **motivational relevance**, sustained attention, and affective processing.

If both occur → an orienting–evaluation sequence.

This is typical for rare emotional stimuli.

The stimulus first triggers orienting (P3a), then emotional evaluation (LPP).

If You Want

I can also provide:

- A full report template for dissociating P3a vs LPP in a paper
- A GLM/design matrix for mass-univariate ERP comparisons
- Expected waveforms for emotional vs neutral novelties
- Guidance for emotionally salient smartphone notification sounds specifically (your earlier dataset)

Just tell me your use case.



Are these (p3a and lpp individually) observed in deviant trials or as difference waves?

Short answer: **Both can be observed directly in deviant trials *and* in difference waves**, but *what you interpret depends on how you compute the comparisons*. Below is a precise explanation useful for EEG/ERP reporting.

1. P3a and LPP Can Be Observed in Raw Deviant-Trial ERPs

P3a

- In oddball paradigms, P3a is typically **clearly visible in the deviant ERP itself**:
A fronto-central positivity around 250–350 ms.
- Even without difference waves, the P3a topography and latency are evident in:
 - Novel stimuli
 - Emotional deviants
 - Rare distractors

LPP

- The LPP is also normally **visible directly in the deviant ERP**:
A sustained **centro-parietal positivity** from ~400–800+ ms.

In other words, ***you do not need difference waves to “see” P3a/LPP.***

Raw ERPs already contain them.

2. But for Interpretation, Difference Waves Are Often Used

Difference waves help isolate the component from overlapping sensory activity.

Typical contrasts

- **P3a: Deviant – Standard**
- **LPP (emotional): Emotional Deviant – Neutral Deviant**
- **Novelty separation: Novel Deviant – Control Deviant (matched acoustically)**

Difference waves remove:

- N1/P2 sensory differences
- Slow drifts
- Physical stimulus differences if control deviants are matched

Why this matters

In raw ERPs, the **P3a** may blend with P3b or sustained P2.

The **LPP** may blend with late slow waves.

Difference waves isolate the *cognitive* effect.

3. Typical Practice in Literature

P3a

- Most studies *report P3a on difference waves* because:
 - Standards have no P3a
 - The subtraction makes the P3a stand out cleanly
 - Overlapping P3b from target trials is avoided

But many papers also show the raw deviant ERP.

LPP

- Emotion studies often use **difference waves**:
 - Emotional – Neutralto isolate emotional processing, not novelty.

However, LPP is also clearly visible in the raw waveforms.

4. Best Practice in Your Case (rare emotionally salient sounds)

For distinguishing novelty P3a vs emotional LPP:

Use two difference waves:

1. Novelty:

- **Neutral Novel Deviant – Standard**

→ isolates P3a (orienting response)

2. Emotion:

- **Emotional Deviant – Neutral Deviant**

→ isolates LPP (affective-motivational processing)

This is the cleanest dissociation.

5. Summary Table

Component	Visible in raw deviant ERP?	Commonly shown as difference wave?	Typical contrast
P3a	Yes	Yes (very common)	Deviant – Standard
LPP	Yes	Yes (common)	Emotional – Neutral

Take-home

- You can see both P3a and LPP in deviant-trial ERPs.
- **Difference waves** are used mainly to *isolate cognitive components* and remove basic sensory activity.
- For emotionally salient deviants, differences waves are **strongly recommended** to cleanly separate P3a vs LPP.

If you'd like, I can lay out an *analysis pipeline or time windows and electrodes* for reliably extracting P3a and LPP from your oddball dataset.



Give me the time windows and electrodes

Thought for 6s >

Nice — here's a compact, practical set of **time windows**, **electrode ROIs**, and **measurement choices** you can use right away to extract **P3a** and **LPP** for rare emotionally-salient sounds. I give recommended defaults plus alternatives if your data look different.

Time windows (recommended)

- **P3a (Novelty P3)** — 220–350 ms post-stimulus
 - Alternative narrower: 240–300 ms (if your data peak earlier)
- **LPP (Late Positive Potential)** — measure as two subwindows:
 - **Early LPP:** 350–600 ms
 - **Late LPP / sustained:** 600–900 ms (extend to 1000 ms if your epoch allows)
 - If you prefer a single window: 350–800 ms.

Electrode ROIs (10–20/typical caps)

Use *clusters* (average across the listed electrodes) rather than a single electrode — this stabilizes estimates.

- **P3a (fronto-central ROI)**
 - Primary cluster: Fz, FCz, Cz
 - Extended cluster (if you have high-density cap): F1, F2, FC1, FC2, Cz
 - Typical reference: average mastoids or common average (report which you used)
- **LPP (centro-parietal / posterior ROI)**
 - Primary cluster: CPz, Pz, POz
 - Extended cluster: CP1, CP2, P1, P2, Pz, PO3, PO4

Measurement choices (recommended)

- **Baseline:** –200 to 0 ms (standard)
- **Filtering:** high-pass 0.1 Hz (or 0.01 Hz if you want very slow waves) and low-pass 30 Hz (or 40 Hz for noisier data). Apply filters before epoching in a reproducible way.

- **Epoch length:** at least –200 to +1000 ms (so you capture LPP late window).
- **Artifact handling:** remove trials with blinks / EMG; consider ICA to remove ocular artifacts.
- **Amplitude metric:** *mean amplitude across the time window and electrode cluster* (preferred for slow, sustained LPP and to reduce noise).
- **Latency metric (if needed):** peak latency within the P3a window or fractional-area latency (e.g., 50% area) — only if SNR is good. Avoid peak latency for LPP (it's sustained).
- **Difference waves / contrasts:** compute (DeviantNeutral – Standard) to isolate P3a; compute (DeviantEmotional – DeviantNeutral) to isolate LPP driven by emotion.

Practical tips & checks

1. **Topographic check:** confirm P3a shows fronto-central maximum around 220–350 ms and LPP shows centro-parietal maximum in 350–900 ms. If maxima swap, reconsider windows.
2. **Habituation check:** plot amplitude across novel trial order — P3a should decay faster than LPP.
3. **Trial counts:** aim for ≥ 20 –30 **clean trials per condition** after artifact rejection for reliable P3a; for LPP (more robust) similar counts but more is better.
4. **Statistical tests:** use cluster-based permutation or mass-univariate tests across time $\times$ electrodes, or run mixed-effects models on mean amplitudes with ROI $\times$ condition factors; correct for multiple comparisons.
5. **Report everything:** windows, electrodes, reference, filter settings, baseline, artifact rejection criteria, and number of trials per condition.

Quick example ROI definitions you can paste into scripts

- P3a_front = ['Fz', 'FCz', 'Cz']
- LPP_post = ['CPz', 'Pz', 'POz']

If your data deviate

- If the P3a peaks earlier (~180–230 ms) shift the P3a window to 180–260 ms.
 - If LPP starts earlier than 350 ms (strong emotional stimuli) you can use 300–800 ms.
- Always justify any deviation with grand-average plots.

If you want, I can:

- produce **example MATLAB / MNE / EEGLAB** code snippets to compute these cluster means, or
- make a small figure showing suggested windows and ROIs using your grand-average waveform (if you upload it). Which would you prefer?



What all can be concluded from p3a?

Here is a **concise but complete, research-grade list** of what you *can* and *cannot* legitimately conclude from the **P3a**, especially in oddball paradigms with rare (including emotional or novel) sounds.

What You *Can* Conclude From the P3a

1. Automatic Orienting of Attention

The strongest and most reliable interpretation:

P3a reflects the brain's involuntary orienting response to unexpected, contextually deviant, or novel events.

It signals that the stimulus triggered:

- Detection of mismatch
- Automatic shifting of attention away from the ongoing task
- Engagement of the frontal–cingulate “salience” network

This holds even when the subject is *instructed to ignore the stimuli*.

2. Stimulus Salience Detection

A larger P3a indicates that the stimulus was:

- More **behaviorally salient**

- More **unexpected**
- More **distinctive** (perceptually or contextually)

This can be physical salience (intensity, change, complexity) or cognitive salience (violates prediction).

3. Activation of the Frontal Attention System

P3a has well-established generators in:

- Anterior cingulate cortex
- Inferior frontal gyrus
- Insula / dorsal salience network

Thus, you can infer **engagement of the frontal orienting system**, not just sensory mismatch.

4. Short-lived Distraction From the Main Task

A robust P3a correlates with:

- Momentary distraction
- Transient interruption of ongoing processing
- Slowed reaction times in concurrent tasks

So P3a can be interpreted as a **neural index of distraction**.

5. Novelty Processing (Novelty P3 subtype)

The P3a is often called the **novelty P3** because it increases when:

- The stimulus is **previously unseen or unheard**
- Stimulus category is new
- Stimuli contain rich, complex, or environmental features

Thus, you can conclude **the brain classified the event as novel**.

6. Arousal + Orienting Interaction (but not pure emotion)

Emotional sounds often elicit both P3a and later LPP.

From the **P3a alone**, you can conclude:

- The emotional sound was more *attention-grabbing*
- There was **emotion-induced orienting**

But the P3a does not reflect deep emotional evaluation — that belongs to LPP.

7. Cognitive Control / Updating of the Model

Modern predictive-processing interpretations propose that the P3a indicates:

- A **prediction-violation signal**
- Brief updating of the brain's model of the environment
- Triggering of attentional reallocation mechanisms

Thus, you can conclude **surprise / prediction error at a cognitive level**.

✗ What You *Cannot* Conclude From the P3a

1. You cannot infer emotional valence or sustained emotion

- P3a = orienting
- LPP = emotion

P3a does not distinguish positive vs negative emotion.

2. You cannot infer conscious processing or awareness

P3a occurs **pre-attentively**, even during:

- Sleep
- Sedation
- Passive viewing
- Distraction tasks

3. You cannot infer task relevance

P3a is strongest for **task-irrelevant** deviants.

If the stimulus becomes task-relevant, P3a decreases and P3b dominates.

4. You cannot interpret P3a as memory encoding or decision-making

Those belong to **P3b** and **LPP**, not P3a.

5. You cannot interpret P3a without considering physical acoustics

If your deviants are louder or more complex, P3a might reflect **physical salience**, not cognitive novelty.

★ Summary: What P3a Means

A fronto-central positive ERP between ~220–350 ms indicates:

“The brain detected an unexpected, salient event and automatically reoriented attention toward it.”

This includes:

- detection of deviance
- cognitive novelty
- automatic distraction
- salience-related prediction error
- engagement of the frontal orienting network

P3a is *not* an emotional, memory, or decision-related component.

If you want, I can give you:

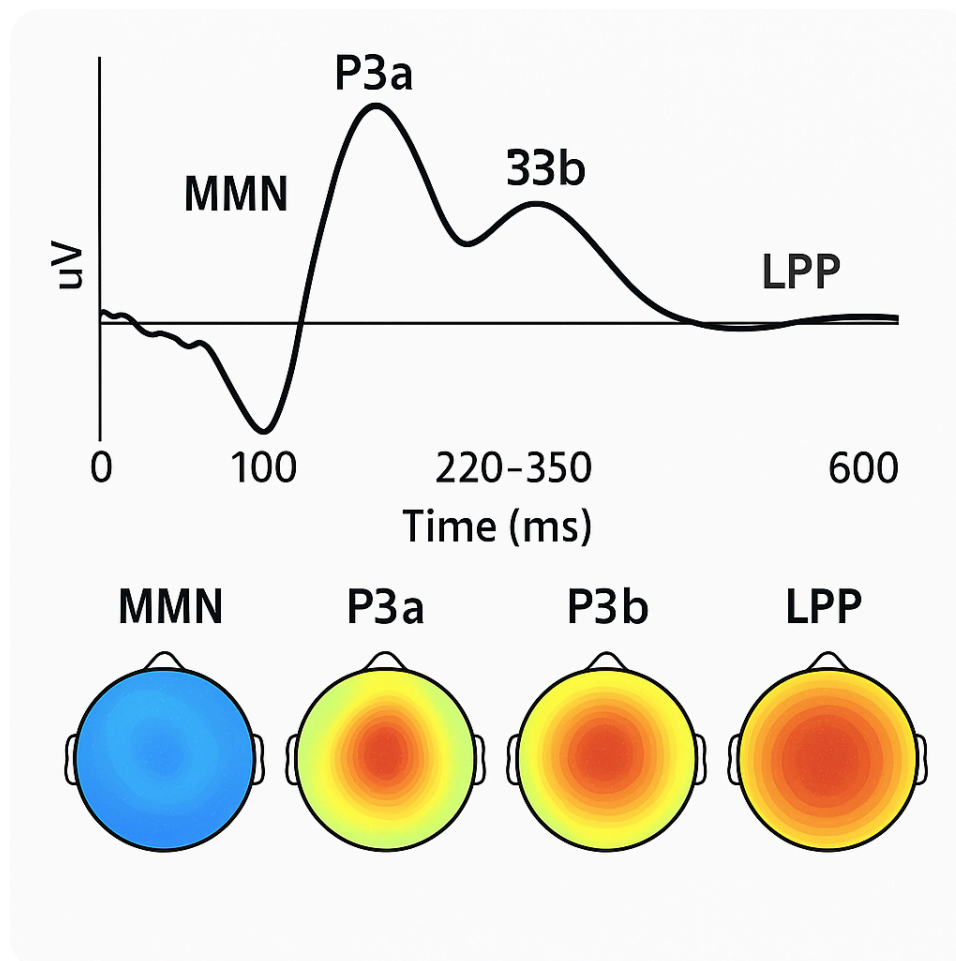
- A one-paragraph “interpretation section” for a paper
- A flowchart: “If P3a increases, what does it mean?”
- A distinction between **P3a vs MMN vs P3b vs LPP** in a single figure

Just tell me what you're writing (discussion, methods, paper, thesis).



Give the distinction in a single figure

Image created >



Distinguish p3a from p2

Here is a **clean, research-grade distinction** between **P3a** and **P2**, focusing on timing, scalp distribution, functional meaning, and how to avoid confusing them.

P3a vs P2 — Key Distinctions

1. Latency (MOST IMPORTANT)

P2

- 150–200 ms post-stimulus

- Often peaks around **170–190 ms**
- Part of early sensory processing

P3a

- **220–350 ms**
- Peaks around **250–300 ms**
- Clearly later than P2

If the positivity is before ~200 ms, it is *not* P3a.

2. Scalp Topography

P2

- **Fronto-central** or **central**
- Usually symmetric
- Often extends slightly to temporal sites
- Topography resembles early perceptual components

P3a

- **Strong fronto-central maximum** (Fz, FCz, Cz)
- More anterior than P3b
- Much broader positivity than P2

P2 = small, early central positivity

P3a = broader fronto-central positivity around 250–300 ms

3. Functional Meaning

P2 → Sensory / categorization

Reflects:

- Early perceptual processing
- Attention-modulated **sensory gain**
- Simple feature evaluation
- Stimulus classification with minimal context involvement

It is *not* linked to novelty or orienting.

P3a → Orienting / salience / novelty

Reflects:

- **Automatic orienting of attention**
- Detection of **novel or unexpected events**
- Early stage of context updating
- Activation of frontal-cingulate salience network

P3a is *context-dependent*; P2 is not.

4. Sensitivity to Stimulus Properties

P2

Strongly affected by:

- Physical intensity
- Spectral complexity
- Onset envelope
- Simple attentional manipulations

P3a

Driven primarily by:

- Contextual novelty
- Mismatch / deviance
- Salience
- Attention capture

If amplitude tracks low-level physical changes → likely P2

If amplitude tracks novelty/surprise → likely P3a

5. Relationship to Deviant Trials

P2

- Standards, deviants, and novels all show P2.

- Difference waves usually show *small* P2 modulations.
- Does NOT dissociate standards vs deviants strongly.

P3a

- Much larger for deviants/novels → hallmark of oddball paradigms.
 - Absent or minimal for standards.
 - Very clear in Difference (Deviant – Standard).
-

6. Shape and Width

P2

- Sharp, narrow peak
- Returns to baseline quickly
- No long tail

P3a

- Broader, rounded peak
- May blend into P3b if the stimulus partially becomes task-relevant

If it's **broad and sustained**, it's not P2.

7. Quick Diagnostic Checklist

Ask yourself these questions:

A. When does it peak?

- < 200 ms → P2
- 220 ms → P3a

B. Is it much bigger for deviants than standards?

- Yes → P3a
- No → P2

C. Is the effect narrow or broad?

- Narrow → P2

- Broad → P3a

D. Does topography shift forward?

- Forward shift (Fz/FCz) → P3a
- Centered slightly posterior → P2

Summary Table

Feature	P2	P3a
Latency	150–200 ms	220–350 ms
Topography	Central / fronto-central	Fronto-central (more anterior)
Function	Early perceptual processing	Orienting; novelty/salience detection
Triggers	Physical features; early attention	Deviance; novelty; unexpected events
Width	Sharp, narrow peak	Broader, larger positivity
Context sensitivity	Low	High
Difference waves	Small	Dominant (deviant–standard)

If you'd like, I can also give you:

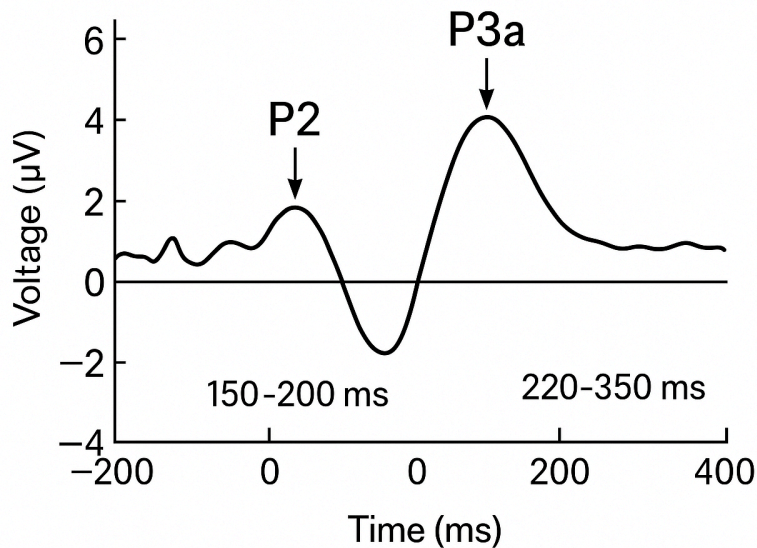
- A simple figure separating P2 and P3a (like the MMN/P3a figure)
- Example scalp maps at peak latencies
- A script for EEGLAB or MNE to extract P2 vs P3a automatically

Just tell me what format you want.



Figure

Image created >



is the negativity between p2 and p3a well-studied? What is it and what does it represent?

Great question — the **negativity between P2 and P3a** *is present in many ERP waveforms*, but it is **not a named component** and is **not typically interpreted as a standalone cognitive process**. Below is the clearest, literature-aligned explanation of what it is and what it is *not*.

✓ Short Answer

No — the negativity between P2 and P3a is *not* a well-studied or well-defined ERP component.

It mostly reflects:

1. The return to baseline after the P2 peak, and

2. The rising slope leading into the P3a.

It does **not** have a unique cognitive interpretation and is **not considered a distinct component**.

✓ Long Answer (Detailed)

1. What is the negativity between P2 and P3a?

It is mostly:

A “trough” created by the overlap of multiple components, including:

- Descending phase of P2
- Rising phase of P3a
- Possibly the tail of N2 or the residual MMN (depending on paradigm)

This trough is often around **180–230 ms**, which is exactly the time when no major, isolated component is expected.

✓ 2. Is it a named component?

No.

It does *not* correspond to:

- N1
- MMN
- N2
- N2b
- N2c
- N3 (a controversial component)

There is **no widely accepted ERP component** that specifically occupies the time between P2 (150–200 ms) and P3a (220–350 ms).

✓ 3. Why does it appear?

Because ERP components sum linearly.

Contributing factors:

- P2 is returning to baseline
- P3a has not yet reached its onset
- Early N2-related activity may be present if the stimulus is a deviant (rare sounds often produce a small N2b or N2-like effect)
- Filtering and baseline choices accentuate the trough

This creates an **artificial “valley”** between two legitimate peaks.

✓ 4. Does it represent cognitive processing?

There is no evidence that it reflects a stable cognitive mechanism.

Even in detailed decompositions (PCA/ICA), the region between P2 and P3a almost always breaks down into:

- P2 component
- N2 component (if present)
- P3a component

The “trough” itself is simply where **none of these components peak**.

✓ 5. Could it be something? (Special cases)

There are a few paradigms in which a small negativity in this window may reflect:

(A) N2b

- Seen for attended deviants
- Peaks ~220 ms
- Fronto-central
- But this occurs only when **participants actively discriminate** or classify sounds

If your experiment is *passive* (e.g., ignoring sounds while watching a movie), N2b is usually **absent**.

(B) Residual MMN tail

- In auditory paradigms, a strong MMN (100–200 ms) can extend into 220 ms

- This can create a shallow negativity right before P3a

(C) The “Pre-P3a trough” (not a real component)

ERP textbooks acknowledge a **consistent trough** before P3a in many paradigms (Picton 2010; Polich 2007), but emphasize that:

“It is not considered an independent component, but rather the transitional region between P2 and the novelty P3.”

— Polich, 2007, *Biological Psychology*

✓ 6. What researchers do with it

They **ignore it**.

No major ERP paper reports or interprets the valley between P2 and P3a, because:

- It doesn’t correspond to a clean cognitive event
- It doesn’t show consistent modulation
- It is highly dependent on the shape of P2 and P3a
- It disappears in many subjects

★ Final Summary

The negativity between P2 and P3a:

- is **not well-studied**
- is **not a named ERP component**
- has **no accepted cognitive interpretation**
- reflects an overlap of P2, N2/MMN tail, and the rising P3a

It is best treated as part of the morphology of the P3a complex, not as a separate component.

If you want, I can also generate:

- A **figure showing how P2, N2, and P3a overlap** to create this trough
- A short **3–4 sentence explanation** for your methods/discussion section